

Pratham Ingale

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Education

Georgia Institute of Technology | Atlanta, GA
Bachelor of Science in Electrical Engineering, GPA: 4.0

Expected Graduation: December 2027

Engineering Experience

SpaceX | Starlink Aviation

September 2025 – December 2025; June 2026 – Present

Avionics Hardware Design Intern (Fall 2025)

Woodinville, WA

- Designed a **60W AC/DC isolated half-bridge LLC resonant converter** using soft switching (ZVS) to **improve efficiency by 15%** over existing flyback. Tuned the hybrid-hysteretic control loop using **AP310** injection to secure **$\geq 50^\circ$ phase margin** and **≥ 20 dB gain margin** across **10–100% load**. Owned schematic & PCB layout; led **bench bring-up**, validating ZVS, resonant-tank behavior, & loop stability.
- Executed staged bring-up of a major **400W** isolated **AC/DC PCBA** design revision across **CrCM PFC, LLC, Boost, & Buck** stages. Validated **PFC current harmonics** to **DO-160** with **PX8000**; characterized efficiency, thermal, & hold-up performance across full load range.
- Led **RCA** for 3 bring-up failures: restored omitted 1V buck-feedback trace (**8.5V runaway**); traced load-switch trips to **chassis-coupled noise**; identified **LLC cross-regulation collapsing** the **auxiliary rail** below converter threshold with main rail unloaded, **rebooting MCU**

Avionics Hardware Design Intern (Summer 2026)

Woodinville, WA

- Increasing AC/DC **input EMI-filter** power capability by **38%, from 400W to 550W**, by identifying magnetic saturation limits, redesigning the filter with replacement components, and **simulating transient** and **frequency-domain** response in **LTspice**; assembling and validating hardware at **full load** for efficiency, thermal performance, and **DO-160 current harmonics**.
- Leading bring-up of radiation-test coupon spanning 8 buck-converter variants; **developed** the full **test plan**, measured **loop gain across load** using **AP310** injection, and **validated efficiency, ripple, protection thresholds**, and startup behavior before radiation testing.
- Leading **RCA** of high-temperature PCBA shutdowns by isolating temperature and load-dependent **diode leakage current** as the failure mechanism; evaluated replacement diodes through board-level rework, **extending operation** from **90s at 90°C** to **> 3hr at 115°C**.

Tesla | Cell Manufacturing

May 2025 – August 2025

Cell Electronics Design Intern

San Diego, CA

- Designed a **4kW, 50V/80A mixed-signal PCB** with **1 Mbps CAN FD**, implementing **discrete window-comparator** circuits for automatic CAN termination, hardware addressing, & connector orientation detection. Integrated pre-charge & load-switch soft-start for **inrush control, OCP, & UV/OV protection**. Designed buck & isolated DC power domains with creepage control & **RLC common-mode filters**.
- Engineered **8-layer, 2oz high-current PCB** rated for **2.5kW (500A @ 5V continuous, 570A peak)**; performed **PDN analysis** to optimize current paths & density, via stitching, & copper geometry; led system bring-up & validated power integrity/thermals at **450A for 700s**.
- Designed a **high-speed digital** star interface for **1 host & 2 nodes** with **<3 ns edges** using **controlled-impedance routing**, ground-via shielding, interplane capacitive coupling, & **damped RLC filtering**; eliminated 12 transceivers while maintaining signal integrity.

Yellowjacket Space Program | Avionics Hardware Lead

August 2024 – Present

Recovery System PCB Designer

Atlanta, GA

- Designed a mixed-signal recovery PCB with **dual-harness battery feeds**, 2x converter paths with **ideal diode OR-ing**, **3x STM32/EKF** channels with SPI IMU/barometer interfaces & **discrete median voting**; added **4x redundant** safing paths (2x RBF, ESTOP, software)
- Executed **6-layer PCB layout** for 4 buck converters and **10 ignition-driver** channels, **minimizing high di/dt hot-loop** and **return-path inductance** while separating high-current switching from low-voltage control circuitry and maintaining redundant-domain isolation.
- Validated design with **simultaneous** ignition of **10 e-matches (300W load step)**, identifying a **downstream transient**; increased 48V to 24V buck **crossover** from **22.6kHz to 49.7kHz**, reducing 3.3V transient from **8.2V to 3.9V** and 24V transient from **32.5V to 26.9V**.

RF Link Evaluation PCB Designer

Atlanta, GA

- Architected a **4-layer 2.4 GHz mixed-signal RF link** PCB for **LoRa transceiver characterization**, combining switched RF test paths, forward & reverse power measurement, board telemetry, and Compute Module 5 control and data logging over **Ethernet and USB**.
- Designed a **TX-power/VSWR measurement** chain for **0–12 dBm transmit power** & **1.1:1–3:1 VSWR** using a **29.2dB** directional coupler and log detector; bounded forward/reverse inputs to **–30 to –17 dBm** and **–56 to –23 dBm** and analyzed **directivity-limited error**.
- Designed **8-channel, 12-bit SAR ADC telemetry** for current, voltage, & temp sensing, including **0–100mA Kelvin-shunt** current sensing, **340Hz anti-alias filters**, & ADC calibration via 1.15V reference; estimated worst-case mux **charge-sharing error** to **~0.10 LSB**.

Skills

Power: Isolated LLC, AC/DC & DC/DC converter design; ZVS/soft switching; loop-gain/stability analysis; EMI filters; PFC & harmonic analysis

Mixed-Signal: I/V/T sensing; SAR ADCs; anti-aliasing/error analysis; comparators & discrete median voters; SPI, I²C, CAN, Ethernet & USB

PCB: PCB stack-up; impedance control & length matching; return-path loop inductance control; creepage; current-density & PDN design

Lab: Oscilloscopes; HV/current probes; e-loads; AP310 FRA; PX8000 power analyzer; IR camera; thermal chambers; power supplies

EDA & Software: Altium Designer, Xpediton Designer, KiCAD, LTspice, SIMPLIS, TINA-TI, Visual Studio Code, GitHub, Enovia, Jira, Excel